

Lesson 1-1: Math is Mine

Connect & apply • 8 tasks • Prepares for Standard MP.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute 30×7 .

Answer: 210

$3 \text{ tens} \times 7 = 21 \text{ tens} = 210$.

Prerequisite skill: Multiply a one-digit number by a multiple of 10

TASK 2 • FOUNDATION 2

Round 47 to the nearest ten, then multiply by 5 to estimate 47×5 .

Answer: 250

47 rounds to 50, and $50 \times 5 = 250$.

Prerequisite skill: Round a two-digit number to the nearest ten to estimate a product

TASK 3 • FOUNDATION 3

A van holds 9 students, and 6 vans go on a trip. Should you add $9 + 6$ or multiply 9×6 to find how many students go? Explain.

Answer: Multiply: there are 6 equal groups of 9, so $9 \times 6 = 54$ students. Adding $9 + 6$ only counts one van and the number of vans.

Compare your explanation with the model answer: check the operation you chose and the value of each part.

Prerequisite skill: Decide when equal groups call for multiplication, not addition

TASK 4 • FOUNDATION 4

Write 64.3 as a tens part plus the rest, then add your parts to check.

Answer: $60 + 4.3 = 64.3$. The 6 is in the tens place, so it is worth 60.

Compare your explanation with the model answer: check the operation you chose and the value of each part.

Prerequisite skill: Decompose a decimal into place-value parts that rebuild it exactly

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TASK 5 • QUICK CHECK A

20×4 ? Estimate 78×6 by rounding 78 to the nearest ten. Is $70 + 8.5$ equal to 78.5?

Answer: $20 \times 4 = 80$. 78 rounds to 80, and $80 \times 6 = 480$, so about 480. Yes: $70 + 8.5 = 78.5$, so it is a correct decomposition.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

30×5 ? Estimate 42×7 by rounding 42 to the nearest ten. Is $60 + 4.2$ equal to 64.2?

Answer: $30 \times 5 = 150$. 42 rounds to 40, and $40 \times 7 = 280$. Yes: $60 + 4.2 = 64.2$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

40×6 ? Estimate 59×4 by rounding 59 to the nearest ten. Is $50 + 3.6$ equal to 53.6?

Answer: $40 \times 6 = 240$. 59 rounds to 60, and $60 \times 4 = 240$. Yes: $50 + 3.6 = 53.6$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A roller coaster has 12 cars that each hold about 6 riders and runs about 9 times an hour. Estimate how many riders it carries in an hour and explain your rounding.

Answer: $12 \times 6 = 72$ riders per run, which is about 70. $70 \times 9 = 630$, so about 630 riders an hour (exactly $72 \times 9 = 648$).

Check the result against the situation and explain your reasoning.

Lesson 1-2: Math is Exploring and Thinking

Connect & apply • 8 tasks • Prepares for Standard 5.NF.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Find $\frac{1}{8}$ of 480.

Answer: 60

Split 480 into 8 equal parts: $480 \div 8 = 60$.

Prerequisite skill: Find a unit fraction of a whole number by dividing

TASK 2 • FOUNDATION 2

Find $\frac{3}{8}$ of 480.

Answer: 180

One eighth is 60, so three eighths is $3 \times 60 = 180$.

Prerequisite skill: Multiply a fraction by a whole number

TASK 3 • FOUNDATION 3

Without calculating, is $\frac{4}{3}$ of 270 more or less than 270? Explain how you know.

Answer: More. $\frac{4}{3}$ is greater than 1, so taking $\frac{4}{3}$ of a number gives more than the whole number ($\frac{4}{3}$ of 270 is 360).

Compare your explanation with the model answer: check the operation you chose and the value of each part.

Prerequisite skill: Predict whether a fraction of a number is smaller or larger than the number

TASK 4 • FOUNDATION 4

Write 105.76 as a hundreds part plus the rest, and add your parts to check.

Answer: $100 + 5.76 = 105.76$. The 1 is in the hundreds place, so it is worth 100.

Compare your explanation with the model answer: check the operation you chose and the value of each part.

Prerequisite skill: Decompose a decimal into place-value parts that rebuild it exactly

Lesson 1-2: Math is Exploring and Thinking

Connect & apply • 8 tasks • Prepares for Standard 5.NF.B.4

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

1/6 of 540? 5/6 of 540? Is 6/5 of 540 more or less than 540?

Answer: $1/6$ of 540 = $540 \div 6 = 90$. $5/6$ of 540 = $90 \times 5 = 450$. More: $6/5$ is greater than 1, so $6/5$ of 540 = 648 is more than 540.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

1/4 of 360? 3/4 of 360? Is 5/4 of 360 more or less than 360?

Answer: $360 \div 4 = 90$. $90 \times 3 = 270$. More: $5/4 > 1$, and $5/4$ of 360 = 450.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

1/10 of 540? 9/10 of 540? Is 2/3 of 540 more or less than 540?

Answer: $540 \div 10 = 54$. $54 \times 9 = 486$. Less: $2/3 < 1$, and $2/3$ of 540 = 360.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A new tower will be $7/6$ as tall as a 540-meter tower. How tall will it be, and how much taller is it than the 540-meter tower?

Answer: $540 \div 6 = 90$, and $90 \times 7 = 630$ meters. It is $630 - 540 = 90$ meters taller, exactly one sixth of 540.

Check the result against the situation and explain your reasoning.

Lesson 1-3: Math is In My World

Connect & apply • 8 tasks • Prepares for Standard 5.NBT.B.7

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute 3.6×100 .

Answer: 360

Shift each digit two places left: $3.6 \times 100 = 360$.

Prerequisite skill: Multiply a decimal by 10 and by 100

TASK 2 • FOUNDATION 2

Compute 7.5×8 .

Answer: 60

$7 \times 8 + 0.5 \times 8 = 56 + 4 = 60$.

Prerequisite skill: Multiply a decimal by a whole number

TASK 3 • FOUNDATION 3

Compute $3,200 \div 80$.

Answer: 40

$320 \text{ tens} \div 8 \text{ tens} = 40$.

Prerequisite skill: Divide a multiple of ten by a two-digit divisor

TASK 4 • FOUNDATION 4

How many minutes are in 6 hours?

Answer: 360

Each hour has 60 minutes: $6 \times 60 = 360$.

Prerequisite skill: Convert hours to minutes

Lesson 1-3: Math is In My World

Connect & apply • 8 tasks • Prepares for Standard 5.NBT.B.7

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

12.5×10 ? 12.5×4 ? $4,000 \div 80$? How many minutes are in 10 hours?

Answer: $12.5 \times 10 = 125$. $12.5 \times 4 = 50$. $4,000 \div 80 = 50$. 10 hours = $10 \times 60 = 600$ minutes.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

8.5×10 ? 8.5×6 ? $3,600 \div 90$? How many minutes are in 5 hours?

Answer: $8.5 \times 10 = 85$. $8.5 \times 6 = 51$. $3,600 \div 90 = 40$. $5 \times 60 = 300$ minutes.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

4.25×100 ? 4.25×8 ? $2,400 \div 60$? How many minutes are in 8 hours?

Answer: $4.25 \times 100 = 425$. $4.25 \times 8 = 34$. $2,400 \div 60 = 40$. $8 \times 60 = 480$ minutes.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A tram carries 80 passengers per ride and each ride takes 12.5 minutes. How many minutes are needed to move 2,000 passengers?

Answer: $2,000 \div 80 = 25$ rides, and $25 \times 12.5 = 312.5$ minutes.

Check the result against the situation and explain your reasoning.

Lesson 1-4: Math is Explaining and Sharing

Connect & apply • 8 tasks • Prepares for Standard 5.MD.C.5

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $5 \times 4 \times 6$.

Answer: 120

$5 \times 4 = 20$, and $20 \times 6 = 120$.

Prerequisite skill: Multiply three whole numbers in any order

TASK 2 • FOUNDATION 2

A box is 7 cm long, 3 cm wide, and 5 cm tall. What is its volume in cubic centimeters?

Answer: 105

$7 \times 3 \times 5 = 105$ cubic centimeters.

Prerequisite skill: Find the volume of a rectangular prism with $V = l \times w \times h$

TASK 3 • FOUNDATION 3

One box holds 240 cubic inches. How many cubic inches do 3 identical boxes hold?

Answer: 720

$240 \times 3 = 720$ cubic inches.

Prerequisite skill: Multiply one box's volume by the number of identical boxes

TASK 4 • FOUNDATION 4

One bed holds 30 cubic feet of soil and another holds 18 cubic feet. How many cubic feet do both hold?

Answer: 48

$30 + 18 = 48$ cubic feet.

Prerequisite skill: Add the volumes of two prisms to find a combined volume

Lesson 1-4: Math is Explaining and Sharing

Connect & apply • 8 tasks • Prepares for Standard 5.MD.C.5

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$8 \times 3 \times 10$? A box is 10 in by 4 in by 10 in: its volume? Two of those boxes hold how much?

Answer: $8 \times 3 \times 10 = 240$. $10 \times 4 \times 10 = 400$ cubic inches. $2 \times 400 = 800$ cubic inches.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$6 \times 2 \times 10$? A box is 5 in by 3 in by 8 in: its volume? Three of those boxes hold how much?

Answer: $6 \times 2 \times 10 = 120$. $5 \times 3 \times 8 = 120$ cubic inches. $3 \times 120 = 360$ cubic inches.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$4 \times 5 \times 6$? A bed is 4 ft by 3 ft by 2 ft: its volume? A second bed holds 24 cubic feet; total?

Answer: $4 \times 5 \times 6 = 120$. $4 \times 3 \times 2 = 24$ cubic feet. $24 + 24 = 48$ cubic feet.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Bed A is 4 ft long, 3 ft wide, and 2 ft deep. Bed B is 6 ft long, 2 ft wide, and 2 ft deep. How much soil fills both beds? Which bed needs more?

Answer: A: $4 \times 3 \times 2 = 24$ cubic feet. B: $6 \times 2 \times 2 = 24$ cubic feet. Together 48 cubic feet; they need the same amount.

Check the result against the situation and explain your reasoning.

Lesson 1-5: Math is Finding Patterns

Connect & apply • 8 tasks • Prepares for Standard 5.OA.B.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Each dime is worth \$0.10. How many dollars are 100 dimes worth? Enter the number only.

Answer: 10

$100 \times 0.10 = 10$, so \$10.00.

Prerequisite skill: Multiply a decimal money value by 100

TASK 2 • FOUNDATION 2

Start at 9 and apply the rule add 4 to write the next four terms.

Answer: 9, 13, 17, 21, 25.

Compare your explanation with the model answer: check the operation you chose and the value of each part.

Prerequisite skill: Extend a pattern by repeatedly applying an add rule

TASK 3 • FOUNDATION 3

A pattern starts at 15 and the rule is add 6. What is the value after 8 steps?

Answer: 63

$15 + 8 \times 6 = 15 + 48 = 63$.

Prerequisite skill: Find a later term as start + number of steps $\times$ rule

TASK 4 • FOUNDATION 4

Which of these could be the number of pennies in a jar worth a whole number of nickels: 132, 145, 158? Explain using the last digit.

Answer: 145, because multiples of 5 end in 0 or 5. 132 and 158 do not end in 0 or 5.

Compare your explanation with the model answer: check the operation you chose and the value of each part.

Prerequisite skill: Recognize multiples of 5 by their last digit

Lesson 1-5: Math is Finding Patterns

Connect & apply • 8 tasks • Prepares for Standard 5.OA.B.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$100 \times \$0.25$? Start at 6 and add 3 four times.
Start at 18 and add 2 ten times.

Answer: $100 \times 0.25 = 25$, so \$25.00. 6, 9, 12, 15, 18. $18 + 10 \times 2 = 38$.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$100 \times \$0.10$? Start at 5 and add 4 three times.
Start at 12 and add 3 ten times.

Answer: $100 \times 0.10 = 10$, so \$10.00. 5, 9, 13, 17. $12 + 10 \times 3 = 42$.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$100 \times \$0.05$? Start at 20 and add 5 four times.
Start at 7 and add 2 eight times.

Answer: $100 \times 0.05 = 5$, so \$5.00. 20, 25, 30, 35, 40. $7 + 8 \times 2 = 23$.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Player A has 38 points and Player B has 36. From now on A scores 2 points per shot and B scores 3 points per shot, one shot each turn. After how many more shots are they tied?

Answer: B gains 1 point on A each turn, and B is 2 behind, so after 2 more shots: $A = 42$, $B = 42$.

Check the result against the situation and explain your reasoning.

Lesson 1-6: Math is Ours

Connect & apply • 8 tasks • Prepares for Standard MP.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

A rack holds 9 bikes with 2 wheels each. How many wheels?

Answer: 18

9 groups of 2: $9 \times 2 = 18$.

Prerequisite skill: Multiply to find the total in equal groups

TASK 2 • FOUNDATION 2

A student says 5 times as many as 12 is 17. Explain the error and give the correct value.

Answer: The student added $5 + 12$. Times as many means 5 copies of 12: $12 \times 5 = 60$.

Compare your explanation with the model answer: check the operation you chose and the value of each part.

Prerequisite skill: Interpret “times as many” as multiplication, not addition

TASK 3 • FOUNDATION 3

Compute 18×4 .

Answer: 72

$10 \times 4 + 8 \times 4 = 40 + 32 = 72$.

Prerequisite skill: Multiply a two-digit number by a one-digit number

TASK 4 • FOUNDATION 4

A rack has 84 wheels and every bike has 2 wheels. How many bikes are on the rack?

Answer: 42

$84 \div 2 = 42$ bikes; check $42 \times 2 = 84$.

Prerequisite skill: Divide to find the number of equal groups

Lesson 1-6: Math is Ours

Connect & apply • 8 tasks • Prepares for Standard MP.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

8×2 ? 6 times as many as 16? How many bikes have 96 wheels if each bike has 2 wheels?

Answer: $8 \times 2 = 16$. $16 \times 6 = 96$. $96 \div 2 = 48$ bikes.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

7×2 ? 4 times as many as 14? How many bikes have 56 wheels if each bike has 2 wheels?

Answer: $7 \times 2 = 14$. $14 \times 4 = 56$. $56 \div 2 = 28$ bikes.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

9×3 ? 5 times as many as 27? How many tricycles have 135 wheels if each has 3 wheels?

Answer: $9 \times 3 = 27$. $27 \times 5 = 135$. $135 \div 3 = 45$ tricycles.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A rack holds 8 bikes with 2 wheels each. A second rack has 6 times as many wheels. How many bikes does the second rack hold, and how many more bikes is that than the first rack?

Answer: $8 \times 2 = 16$ wheels; $16 \times 6 = 96$ wheels; $96 \div 2 = 48$ bikes. That is $48 - 8 = 40$ more bikes.

Check the result against the situation and explain your reasoning.